



Technical Assessment Test

General Directions

Complete the following puzzle in as a web application. Feel free to use front-end framework of your choice (JavaScript or Angular or React, etc.) and style framework of your choice (bootstrap, angular material, etc.)

Note: Your solution needs to be considered as production ready for deployment.

The Problem

Problem 1

You are required to build a **Simple Task Management Portal** intended for internal users.

The portal must include a **login screen** to authenticate users. Upon successful login, the user should be redirected to a **dashboard**. The dashboard design is intentionally open-ended—candidates are encouraged to propose their own layout and user experience, as long as it is intuitive and user-friendly.

The application must also provide functionality to:

- View a list of all tasks
- View the details of a single task
- Edit an existing task

The overall solution should demonstrate good front-end design principles, usability, and responsiveness across web and mobile devices

Problem 2

Design the high-level architecture for a modern web app created in the technical assessment. Provide an architecture diagram showing:

- UI composition approach (SPA/MPA, micro-frontends or not)
- State management strategy

- API interaction pattern
- Authentication/authorization flow
- Error handling, logging, analytics
- Build/deploy pipeline (at high level)

User Stories & Acceptance Criteria (Given / When / Then)

User Story 1: Login Screen

As a user

I want to log in to the application

So that I can access the system securely

Acceptance Criteria

Given I am on the login page

When I enter a valid username and password

Then I should be logged in successfully and redirected to the dashboard

Given I enter invalid credentials

When I click the login button

Then I should see a validation error message

☒ Must include:

- Form validation (required fields)
- Error message or toast on failure
- Mobile-friendly layout

User Story 2: Responsive Dashboard Layout

As a user

I want the application to work well on mobile and desktop

So that I can use it on any device

Acceptance Criteria

Given I open the application on desktop or mobile

When the screen size changes

Then the layout should adapt appropriately without breaking

☒ Must include:

- Responsive navigation or layout
 - Proper spacing and readability on mobile
-

User Story 3: Task List with Table View

As a user

I want to view a list of tasks

So that I can easily see available information

Acceptance Criteria

Given I am on the task list page

When the page loads

Then a table of tasks should be displayed

☒ Table must include:

- Task Name
- Status
- Assigned To
- Action column (View / Edit)

☒ Bonus:

- Sorting or basic filtering
-

User Story 4: Toast Notification

As a user

I want to receive feedback when actions are completed

So that I know the result of my actions

Acceptance Criteria

Given I create or update a task

When the action is successful

Then a toast notification should be displayed

☒ Toast types:

- Success
 - Error (optional)
-

User Story 5: Task Creation Form with Validation

As a user

I want to create a task using a form
So that I can add new data into the system

Acceptance Criteria

Given I am on the create task page

When I submit the form with missing or invalid fields

Then validation messages should be shown

☒ Validation examples:

- Required fields
 - Minimum character length
-

User Story 6: Image Display**As a user**

I want to see an image related to a task or profile
So that the UI feels more visual and user-friendly

Acceptance Criteria

Given I am viewing a task or profile page

When the page loads

Then an image should be displayed correctly and responsively

☒ Image can be:

- User avatar
 - Task-related placeholder image
-

User Story 7: Collapsible View**As a user**

I want to expand and collapse sections
So that I can focus only on relevant information

Acceptance Criteria

Given I am on the task detail page

When I click on a section header

Then the content should expand or collapse

☒ Example:

- Task details section

- Additional notes section
-

User Story 8: Step View (Wizard)

As a user

I want to complete a task creation process in steps
So that complex input is easier to manage

Acceptance Criteria

Given I am creating a new task

When I move through the steps

Then the system should guide me step by step

☒ Step examples:

1. Task Information
 2. Assignment Details
 3. Review & Submit
-

Non-Functional Requirements

- Clean, readable, and well-structured code
 - Reusable components
 - Clear folder structure
 - Meaningful variable and function names
 - Basic error handling
 - No hardcoded UI-breaking values
-

Sample JSON Payload

```
{
  "auth": {
    "users": [
      {
        "username": "admin",
        "password": "Admin@123",
        "displayName": "Admin User",
```

```
    "avatarUrl": "https://via.placeholder.com/150"
  }
]
},
"uiConfig": {
  "toasts": {
    "loginSuccess": "Login successful",
    "loginError": "Invalid username or password",
    "taskCreated": "Task created successfully",
    "validationError": "Please fix validation errors"
  },
  "steps": [
    "Task Information",
    "Assignment Details",
    "Review & Submit"
  ]
},
"tasks": [
  {
    "id": 1,
    "name": "Design Login Screen",
    "status": "In Progress",
    "assignedTo": "John Doe",
    "priority": "High",
    "description": "Create responsive login screen with validation",
    "imageUrl": "https://via.placeholder.com/300x200",
    "details": {
      "createdDate": "2026-03-01",
      "dueDate": "2026-03-10",
      "notes": "Ensure mobile-first design"
    }
  },
  {
    "id": 2,
    "name": "Implement Task Table",
    "status": "Completed",
    "assignedTo": "Jane Smith",
    "priority": "Medium",
    "description": "Display task list in table view",
    "imageUrl": "https://via.placeholder.com/300x200",
    "details": {
      "createdDate": "2026-03-02",
      "dueDate": "2026-03-08",
      "notes": "Add sorting if possible"
    }
  }
],
```

```
"formSchema": {
  "createTask": {
    "fields": [
      {
        "name": "taskName",
        "label": "Task Name",
        "type": "text",
        "required": true,
        "minLength": 3
      },
      {
        "name": "assignedTo",
        "label": "Assigned To",
        "type": "text",
        "required": true
      },
      {
        "name": "priority",
        "label": "Priority",
        "type": "select",
        "options": ["Low", "Medium", "High"],
        "required": true
      }
    ]
  }
}
```

☒ How Candidates Are Expected to Use It (High-Level)

- Mock login by validating against auth.users
- Render task list in a **table**
- Use details for **collapsible sections**
- Display imageUrl responsively
- Show toast messages from uiConfig.toasts
- Drive stepper UI using uiConfig.steps
- Validate forms using formSchema

Code Submission Instructions

create a **PRIVATE** GitHub repository and share it with the user “**cityfmcodetests**”. Also provide readme file on how to run the application on local.